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Pretangle tau pathology accrual in the default mode network during the progression of Alzheimer's disease

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Abstract

Alzheimer's disease (AD) is a secondary tauopathy characterized by the accumulation of aggregated amyloid-beta (Aβ) peptides as extracellular neuritic and diffuse plaques, alongside hyperphosphorylated tau aggregates forming intracellular neurofibrillary tangles (NFTs). Although NFT pathology is strongly linked to neurodegeneration and cognitive decline in AD, growing evidence suggests that soluble "pretangle" aggregates of tau, formed prior to NFT development, are the primary neurotoxic species driving disease progression and cognitive impairment. We quantified pretangle tau moieties in postmortem tissues of the default mode network (DMN), a resting state connectome that mediates episodic memory and falters early in AD, to determine the association between DMN tau pathology and antemortem cognitive status. To accomplish this, we assayed postmortem fixed and frozen tissue from frontal cortex (FC), posterior cingulate cortex (PCC), and precuneus (PreC) DMN hubs obtained from participants of the Rush Religious Orders Study (RROS). Immunohistochemical and biochemical quantification was performed using site-specific tau antibodies recognizing pathological pretangle epitopes pS422 (phosphorylation at tau residue serine 422), TOC1 (tau oligomers), and TNT2 (N-terminal tau misfolding), as well as the mid-stage NFT marker TauC3 (C-terminal tau truncation). Pretangle tau profiles included neuropil threads and neuronal inclusions as early as Braak stage III, with significant increases from Braak stage IV to V across the DMN hubs. There was a step-wise increase in the pretangle markers in subjects who died with no cognitive impairment (NCI) to mild cognitive impairment (MCI) to AD. The increase in pretangle tau pathology accrual correlated with poorer antemortem neuropsychological tests, including episodic memory, semantic memory, and a global cognitive *z* score calculated for each RROS participant. Pretangle tau accrual also correlated with postmortem neuropathological diagnostic measures of AD severity. In addition, we found that the PCC displayed greater tau pathology compared to the PreC, suggesting selective vulnerability within DMN hubs. Collectively, our findings are a first of its kind cellular and protein quantitative analysis to demonstrate neurotoxic pre-tangle tau accumulation within the DMN connectome that highly correlated with cognitive decline in MCI and AD, underscoring pretangle tau as a potential early driver of disease progression and a target for therapeutic intervention.

Keywords Tau, Alzheimer's disease, Default mode network, Cognition, Pretangle, Oligomer

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Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative dementing disorder and the fifth leading cause of death among those aged 65 and older [1]. The postmortem presence of amyloid beta (Aβ) peptide-containing plaques and neurofibrillary tangles (NFTs) composed of aberrantly post-translationally modified tau proteins defines AD pathologically. Tau is an intrinsically disordered protein that plays various roles in cellular function in its normal physiological state [2]. However, when aberrant post-translational modifications accumulate, tau undergoes conformational changes that lead to self-aggregation and fibrillization, ultimately coalescing as NFTs in AD and other tauopathies. In AD, NFTs accumulate in the transentorhinal cortex and hippocampus prior to spreading to limbic and neocortical regions, thus guiding the Braak staging system [3]. Historically, research on tau pathology in AD has focused on NFTs as the primary source of toxicity and neuronal death. By contrast, injections of recombinant tau fibrils into the hippocampus did not induce neuron loss or synaptic dysfunction in wild type mice [4], and two-photon imaging in transgenic tau mice revealed that NFT-bearing cortical neurons respond to sensory inputs and remain integrated in local circuits [5]. Whereas turning off tau expression in a transgenic tau mouse model rescued memory and prevented neuronal loss, even as NFT-like inclusions continued to form [6]. Since these data suggest that frank NFTs are less toxic to neuronal function, research has increasingly shifted towards investigating pretangle tau aggregates that arise from early aberrant phosphorylation, protein misfolding events, and oligomer formation.

In contrast to NFTs, soluble tau oligomers can seed tau pathology [7], disrupt mitochondrial homeostasis through impaired energy metabolism, increased reactive oxygen species, and calcium imbalances [4, 8–12], interfere with bidirectional axonal trafficking and microtubule assembly [9, 13–15], and diminish synaptic vesicular signaling and long-term potentiation (LTP) [9, 14, 16–19]. These oligomeric tau insults are postulated to culminate in a toxic cellular environment promoting neurodegeneration, synaptic loss, disconnection, and cognitive decline.

Clinically, cognitive decline in episodic and semantic memory domains is an early phenomenon in AD progression [20–22]. However, the relationship between pathological tau accrual and memory dysfunction might not be linear due to individual differences in cognitive reserve and AD-associated risk factors [23]. Increasing tau burden (measured by PET imaging) correlates with reduced resting-state functional connectivity (measured by functional MRI (fMRI)) and cognitive deterioration [24–32]. Most studies on tau pathology focus on the medial temporal lobe due to its crucial role in episodic memory coding and retrieval [33–35]. However, the default mode network (DMN), a resting-state network that is active by default in the absence of an attention-demanding task, is a key memory connetome that requires investigation to understand the role tau pathology plays in mild cognitive impairment (MCI/prodromal AD) prior to the advanced clinical representation of AD [36].

The DMN consists minimally of three hubs: frontal cortex (FC), posterior cingulate cortex (PCC), and precuneus (PreC) [37, 38]. The onset of Aβ-PET positivity within the DMN is associated with signs of hyperexcitability [39] and hyperconnectivity [40–42]. By contrast, during the onset of tau-PET positivity, DMN functional connectivity shifts towards hypoconnectivity [40], in addition to delayed task deactivation in amnestic MCI patients. [43, 44]. Notably, reductions in DMN connectivity were observed in individuals experiencing subjective cognitive decline (SCD) [41] or newly diagnosed with MCI [45]. Previous studies that combined fMRI and tau PET imaging revealed that tau deposition in DMN regions correlates with aberrant network connectivity [46]. Other studies combining CSF tau and fMRI demonstrated associations between DMN connectivity and CSF ptau levels, suggesting that the accumulation of ptau impairs DMN connectivity [36]. However, imaging and CFS biomarker studies do not elucidate the role that pretangle tau pathology plays in cortical cellular degeneration and protein levels within hubs of the DMN during the progression of AD, which remains a significant knowledge gap. Currently, there is an absence of pretangle-specific PET ligands needed to track prefibrillar tau accumulation in living patients. Moreover, cross-sectional epitope-specific detection of soluble pretangle tau within DMN hubs in subjects with varying antemortem cognitive function and postmortem neuropathological diagnostic assessments has yet to be explored.

This study aimed to quantify pretangle tau in postmortem DMN regions (FC, PCC, and PreC) obtained from Rush Religious Orders Study (RROS) subjects who died with a clinical diagnosis of no cognitive impairment (NCI), MCI, or AD that displayed a full range of Braak NFT stages. Pretangle pathology was identified using epitope-specific tau antibodies that recognize early post-translational modifications related to soluble tau aggregation and toxicity combined with quantitative morphometrics and custom sandwich ELISA analysis of soluble brain fractions. We also tested for associations between the abundance of specific pretangle markers within each DMN region and antemortem neuropsychological test performance and postmortem neuropathological diagnostic criteria. Given the current lack

of pretangle tau-specific in vivo probes, this is the first comprehensive analysis of pretangle tau accrual within the DMN during the progression of AD in a well-characterized, clinical pathologic cohort.

Materials and methods

Subjects and clinical pathologic assessments

Case-matched fixed and frozen FC, PCC, and PreC tissue samples were obtained from the RROS, a longitudinal clinical pathological study of aging and dementia in elderly Catholic clergy [47–49] provided to our ongoing Program Project grant (AG014449). Subjects were classified antemortem as NCI (n=47), MCI (n=36), or AD (n=36). Details of RROS clinical and neuropathologic evaluations and diagnostic criteria are published extensively [47, 48, 50, 51]. Briefly, RROS participants undergo an annual neurological examination and a battery of cognitive performance tests that includes the Mini-Mental State Exam (MMSE) and 19 additional neuropsychological tests of five cognitive domains: orientation, attention, memory, language, and perception [48]. Composite scores of episodic memory, semantic memory, working memory, perceptual speed, and visuospatial ability, as well as a composite global cognitive z-score (GCS), were derived from this test battery for each subject within a year of death (Supplementary Table 1). NCI subjects did not exhibit impairment in any of these domains after adjusting for age and education [48, 52]. The MCI subjects exhibited impairment in single or multiple cognitive domains after adjusting for age and education, but did not meet the criteria for AD or dementia [48, 52], which is consistent with criteria used by others in the field [53–55]. The diagnosis of dementia due to AD followed the revised recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for AD [56]. Exclusion criteria included a history of major depressive disorder, chronic alcoholism, and/or neuropathological evidence of Lewy body diseases, extensive TDP-43 proteinopathy, hippocampal sclerosis, and/or large strokes. Apolipoprotein E (APOE) genotyping was performed as reported previously [57].

For postmortem neuropathological analysis, tissue immersion fixed in 4% paraformaldehyde was embedded in paraffin, sectioned at 8 μm and immunostained with antibodies against Aβ (6E10, 1:400 dilution) and phosphorylated tau (AT8, 1:250 dilution). A board-certified neuropathologist evaluated all cases while blinded to clinical diagnosis [51]. Designations of "normal" (with respect to AD or other dementing processes), "possible AD", "probable AD", or "definite AD" were based on semi-quantitative estimation of neuritic plaque density, an age-related senile plaque score, and the presence or absence of dementia as established by the Consortium to Establish a Registry for Alzheimer's Disease (CERAD) [58]. Braak scores for the staging of NFT pathology were established for each case [3]. Cases also received an NIA-Reagan Likelihood-of-AD diagnosis based on neuritic plaque and NFT pathology [59]. Semi-quantitative estimates of global NFT, neuritic plaque, and diffuse plaque burden were also derived from summary evaluations of the entorhinal cortex, hippocampus, and midtemporal, inferior parietal, and midfrontal cortices [52]. The "ABC" algorithm for the diagnosis of AD [60] was applied to all cases.

For validation studies, frozen FC, PCC, and PreC tissue blocks from low-Braak/nondemented aged controls (Braak stages I-II), mid-Braak (Braak stages III-IV), and high-Braak (Braak stages V-VI) (n=12/group, n=36 total) subjects were obtained postmortem from Michigan AD Center (MADC) clinical cohort [61]. These subjects underwent routine neuropathological diagnostic analysis [61]. MMSE scores were utilized to calculate correlations between pre-tangle tau pathology and global cognition.

Immunohistochemistry

Forty micron-thick free-floating fixed tissue sections from each DMN region were washed in 0.1%Triton X-100 in TBS (TBS-T) buffer, quenched with peroxidase (3% in methanol), blocked using 10% NGS/2% bovine serum albumin in 1xTBS-T, and incubated with the following primary antibodies (Table 1), alone (for brightfield microscopy) or in combination (for confocal microscopy): 1) pS422 rabbit monoclonal antibody (Millipore AB79415, 1:2,500), which recognizes soluble tau phosphorylation at the serine 422 residue, an early event in NFT evolution [62, 63] that facilitates stable tau dimerization and impairs axonal transport [13]; 2) TOC1 mouse monoclonal IgM antibody (1:20,000; RRID: AB_2832939), which recognizes tau oligomers by binding to a continuous epitope within the proline-rich region (amino acids 209–224) formed by tau dimer antigen [64, 65]; or 3) TNT2 mouse monoclonal IgG1 antibody (1:200,000; RRID: AB_2736931), which recognizes the exposed phosphatase-activating domain (PAD) in the N-terminus as an early conformational change linked to axonal transport impairment [66]. Neither TOC1 nor TNT2 epitopes colocalize with thiazine red in neurons displaying pretangle stages of tau accumulation [64, 66]. In addition, sections were also immunostained using TauC3 mouse monoclonal antibody (1:6,000; RRID: AB_2832930), which recognizes a truncation neoepitope at D421 in the C-terminus and serves as a marker for mid-stage NFT development [67, 68]. Finally, the PHF-1 mouse monoclonal antibody (1:20,000; RRID: AB_2315150), which detects phosphorylated serine residues at positions 396 and 404, was used to detect filamentous tau and NFTs [69]. In parallel experiments, we

Table 1 Antibodies and working concentrations

Antibody	Working concentrations	Isotype	Epitopes
Ps422	IHC (1:2500) IF (1:500) ELISA (1:500)	Rabbit polyclonal	pS422
TOCI	IHC (1:20,000) IF (1:2,000) ELISA (1:500)	Mouse IgM	Amino acids 209–224
TNT2	IHC (1:200,000) IF (1:20,000) ELISA (1:100)	Mouse IgG1	Amino acids 2–18 (Phosphatase-activating domain)
TauC3	IHC (1:6,000) IF (1:500) ELISA (1:500)	Mouse IgG1	Amino acid 421 (C-terminus truncation)
PHF-1	IHC (1:20,000)	Mouse IgG1	Ps396/Ps404
Tau5	ELISA (1:1000)	Mouse IgG1	Amino acids 218–225
MOAB-2	IHC (1:5,000)	Mouse IgG2b	Amino acids 1–4

used the MOAB-2 mouse monoclonal antibody (1:5,000; RRID: AB_2313888), which detects oligomeric Aβ and plaque pathology via binding residues 1–4 within the N terminus [70].

For brightfield immunohistochemistry (IHC), sections underwent three 10-min TBS-T washes before the secondary incubation with either mouse or rabbit biotinylated secondary antibodies (1:500) for one hour at room temperature. After applying the ABC kit (Vector Laboratories PK-6100) for one hour and three more 10-min TBS-T washes, 3,3′-Diaminobenzidine (Vector Laboratories SK-4105) was used to develop brown reaction products. Sections were then washed, mounted on slides, counterstained with cresyl violet (Sigma Aldrich C5042) and coverslipped for digital imaging as described below.

For immunofluorescence (IF) , sections were first exposed to a light-emitting diode (LED) light source for 16 h to quench autofluorescence. On Day 1, the sections were blocked with normal goat serum and incubated overnight with the TNT2 primary antibody (1:20,000) or TauC3 (1:2,500). The following day, sections were washed and incubated with Alexa568 anti-mouse IgG1-specific secondary antibody (1:500, Invitrogen A21124) for one hour. Before adding the other two primary antibodies TOC1 (1:5,000) and pS422 (1:2,000), the sections were treated with mouse Fab fragments (1:20, Jackson Laboratories, AB-2338476) for one hour to prevent non-specific binding between the two mouse antibodies. On Day 3, sections were washed again and incubated with Alexa-488 anti-mouse IgM-specific (Invitrogen, A21042, 1:500) and Alexa-405 anti-rabbit IgG (Invitrogen, A31556, 1:500) secondary antibodies for one hour before mounting and coverslipping. Sections were imaged using a Nikon A1 laser scanning confocal microscope with a 40× oil immersion objective.

Whole slide digital imaging and quantitative analysis

Tissue sections were digitally scanned using a Zeiss AxioScan7 with a TL LED lamp and a 20x/0.8 M27 plan-apochromatic objective. The images were then stitched into a single composite using Zen Blue software (version 3.7.97.07000) with a pixel resolution of 0.173 µm × 0.173 µm. Five sections per case per region and per antibody were used for quantification. The entire gray matter was analyzed as the region of interest. After annotations were drawn, thresholds were adjusted for each antibody, and HALO (Area Quantification v2.1.11, Indica Labs) machine learning software was used to quantify the signal intensity and tissue area associated with the signal. For normalization purposes, the percentage of tissue area stained was used as the outcome measure.

Soluble tau extraction

Frozen tissue (150 mg) from RROS and MADC cases was placed in a 10X volume of brain homogenization buffer (50 mM Tris, 275 mM NaCl, 5 mM KCl) supplemented with protease inhibitors (10 µg/ml of Pepstatin A, Leupeptin, Bestatin, Aprotinin, and 1 mM PMSF). The tissue was homogenized on ice using a Tissue Tearor rotor–stator probe (Biospec) and centrifuged at 27,000 × *g* for 20 min at 4 °C to obtain the soluble pretangle tau-enriched fraction for custom sandwich ELISAs. Protein concentrations were determined using the BCA method (Thermo Fisher Scientific, A53226), and all samples were stored at −80 °C.

Custom sandwich ELISAs

Custom sandwich sELISAs were prepared using TOC1 (2 µg/ml), TNT2 (1 µg/ml), TauC3 (2 µg/ml), and Tau5 (total tau, 1 µg/ml) antibodies as capture antibodies [71–73],

combined with R1 (1:10,000; RRID: AB_2832929), a pan-tau antibody, for detection [74]. Briefly, 96-well plates were coated with the capture antibodies in borate buffer (100 mM borate acid, 25 mM sodium borate, 75 mM NaCl, 0.25 mM thimerosal) and incubated for one hour at room temperature on a shaker. After two quick washes with wash buffer (100 mM borate acid, 25 mM sodium borate, 75 mM NaCl, 0.25 mM thimerosal, 0.4% (w/v) bovine serum albumin, 0.05% (v/v) Tween-20), the wells were blocked with 5% non-fat dry milk for one hour. Brain lysate samples were then added to the wells and incubated on the shaker for 90 min using predetermined concentrations from previous titration experiments (Samples: TOC1&TauC3: 20ug/well, TNT2&Tau5:10ug/well). Each assay included recombinant tau monomer and aggregate standards (1:3 serial dilution starting from: TOC1:200 nM, TNT2: 100 nM, Tauc3: 100 nM, Tau5:55Nm) generated as previously described [66, 75, 76] to estimate levels of tau species in the lysate samples. Following four additional washes, the R1 detection antibody was added for a 90-min incubation, followed by a one-hour incubation with anti-rabbit horseradish peroxidase (HRP)-conjugated secondary antibody (1:5,000). TMB solution (Sigma, T0440) was then added for color development, allowing the reaction to proceed for 4–8 min before terminating with the stop buffer (3.5% sulfuric acid). Finally, plates were read with a plate reader at 450 nm. The readout was transformed into the % light absorbed with the Y = 100-(100*10^(-Y)) formula (Y = absorbance).

Statistical analysis

All datasets were subjected to the Shapiro–Wilk test for normality. Subject demographic, clinical, and pathological variables were compared by either one-way ANOVA or the Kruskal–Wallis test with post hoc corrections for multiple comparisons. Chi-Square testing was used to compare sex and *APOE* ε4 allele distribution across the groups. Tau marker levels were compared within each region across clinical diagnosis or Braak stage using the Kruskal–Wallis test with Dunn's correction. The non-parametric Friedman test with Dunn's correction was applied for interregional comparisons of markers as paired data. Spearman rank correlations were used to test for associations between regional tau levels and antemortem neuropsychological test scores or postmortem variables. Significance was set at $\alpha = 0.05$.

Results

RROS cohort demographics and clinical neuropathologic characteristics

Detailed demographic, clinical, and neuropathological data for the RROS participants ($n = 119$) are listed in Table 2. No significant differences were observed in sex, years of education, or postmortem interval (PMI) among the NCI, MCI, and AD clinical groups. APOE ε4 allele was present in 21.8% of the subjects, but its prevalence did not differ across groups. Age was significantly lower in the NCI compared to the MCI and AD subjects ($p = 0.0002$). However, age was not correlated with select tau marker levels in most of the DMN regions, with the exception of TOC1 in the FC and TNT2 in the PreC in the sELISA analyses (Supplementary Table 2). Global cognitive performance on both the MMSE and GCS was significantly lower in the MCI compared to NCI group ($p < 0.0001$); these scores were also significantly lower in AD compared to MCI ($p < 0.0001$). Braak, CERAD, and Reagan neuropathological diagnostic scores were significantly more severe in AD compared to NCI and MCI (Table 2).

Pathological tau accrual in the DMN in the RROS cohort

Fixed free-floating FC, PCC, and PreC tissue sections (NCI = 19, MCI = 16, AD = 16, n = 51 total) were labeled separately with the select site-specific tau antibodies (pS422, TOC1, TNT2, and TauC3) for IHC quantification. PHF-1, a canonical NFT marker, was used in the PCC samples to compare with the other markers listed above. As shown in Fig. 1, pretangle markers of tau pathology were observed only sparsely as neuropil threads (NTs) in Braak stages III/IV, with an appreciable accumulation of NTs and the appearance of NFTs within cortical layers 2/3 and 5/6 by Braak stage V. Qualitatively, pS422 + NTs appeared to accumulate prior to the TOC1 + and TNT2 + NTs, whereas TauC3 appeared more confined to NFTs in later Braak stages (Fig. 1). Multiplexed fluorescence microscopy of the pS422, TOC1, and TNT2 pretangle markers in higher Braak cases revealed co-labeling in the cell bodies of both pretangle and NFT-bearing neurons, as well as within NTs (Fig. 2) and neuritic plaques (Supplementary Fig. 1 A). Interestingly, different distribution patterns of the markers were noted within individual NTs (Fig. 2, Supplementary Fig. 1 A). The mid-stage marker TauC3 [63] did not significantly overlap with earlier pretangle markers within cell bodies, but dual labeled with NTs immunostained by the other pretangle markers (Supplementary Fig. 1B).

Quantitative morphometrics of the brightfield immunolabeled sections revealed that pS422 + profile levels were significantly increased from Braak stage IV to V in all three regions (FC, $p = 0.0245$; PCC, $p = 0.0214$; PreC, $p = 0.0111$) (Fig. 3A, C, E; Supplementary Fig. 2A). TOC1 levels also began to rise between Braak stages IV and V in all three DMN hubs but reached statistical significance only at Braak stage VI (FC, $p = 0.025$; PCC, $p = 0.0154$, PreC, $p = 0.0171$, Fig. 3A, C, E; Supplementary Fig. 2 A). TNT2 labeling exhibited a significant increase by Braak stage V in FC and PCC ($p = 0.0351$

Table 2 Demographic, clinical, and pathological profile of the RROS cohort

	Clinical diagnosis				Comparison by diagnosis group
	NCI (*n* = 47)	MCI (*n* = 36)	AD (*n* = 36)	Total (*n* = 119)	(*P* value)
Age at death(years)					
Mean ± SD	88.1 ± 5.1	92.2 ± 5.9	92.3 ± 4.5	90.6 ± 5.5	0.0003[a]
(Range)	(70.5, 98.1)	(80.7,103.7)	(80.7,102.1)	(70.5,103.7)	
No. (%) Males	18 (37.5%)	13 (36.1%)	7 (19.4%)	38 (31.7%)	0.1535[‡]
Years of Education					
Mean ± SD	18.6 ± 3.3	18.0 ± 2.8	17.6 ± 3.3	18.1 ± 3.2	0.2943[a]
(Range)	(10, 27)	(12, 24)	(8, 27)	(8, 27)	
Postmortem Interval (hours)					
Mean ± SD	11.5 ± 6.5	9.8 ± 6.7	9.2 ± 7.1	10.3 ± 6.8	0.1062[a]
(Range)	(3.2, 27.3)	(0.9, 24.0)	(2.7, 34.3)	(0.9, 34.3)	
No. (%) ApoE4 (≥ 1 allele)	11 (23.4%)	7 (19.4%)	8 (22.2%)	26 (21.8%)	0.9023[‡]
MMSE					
Mean ± SD	28.0 ± 1.8	24.2 ± 3.2	16.3 ± 6.6	23.4 ± 6.4	< 0.0001[a]
(Range)	(21, 30)	(19, 30)	(0, 26)	(0, 30)	
Global Cognitive Score					
Mean ± SD	0.2 ± 0.4	−0.6 ± 0.4	−2.1 ± 0.9	−0.7 ± 1.1	< 0.0001[b]
(Range)	(−0.7, 1.1)	(−1.6, 0.2)	(−4.1, −0.4)	(−4.1, 1.1)	
Braak Scores					
I-II	14	5	2	21	< 0.0001[a]
III-IV	31	25	11	67	
V-VI	2	6	23	31	
NIA Reagan Diagnosis (likelihood of AD)					
No AD	0	0	0	0	< 0.0001[a]
Low	30	16	5	51	
Intermediate	16	15	15	46	
High	1	5	16	22	
CERAD Diagnosis					
No AD	18	11	3	32	0.0003[a]
Low	4	3	1	8	
Intermediate	18	12	14	44	
High	7	10	18	35	

[a] Kruskal–Wallis and Dunn's correction

[b] One-way ANOVA and Tukey's correction

[‡] Chi-Square test Abbreviations: *AD* Alzheimer's disease, *MCI* Mild cognitive impairment, *MMSE* Mini-Mental State Examination, *NCI* No cognitive impairment, *SD* Standard deviation

and p = 0.0003, respectively) (Fig. 3A, C, E; Supplementary Fig. 2 A). TauC3, which displayed a lower abundance than the pretangle markers, was significantly increased in Braak V in the FC and Braak VI in PCC (p = 0.0176 and p = 0.0477, respectively, Fig. 3A, C, E; Supplementary Fig. 2A). By contrast, MOAB-2 immunoreactivity revealed increases in amyloid pathology as early as Braak stages II-III, but the increase was significant only in the PreC in Braak stage VI (PreC, p = 0.0144, Fig. 3A, C, E; Supplementary Fig. 2 A).

When compared among the clinical diagnostic groups, pS422 pathology was significantly increased between NCI and AD in all three regions (FC, p = 0.0245; PCC, p = 0.0049; PreC, p = 0.0042), with increased levels in AD compared to MCI in FC and PCC but not PreC (Supplementary Fig. 3A). TOC1 levels were also significantly

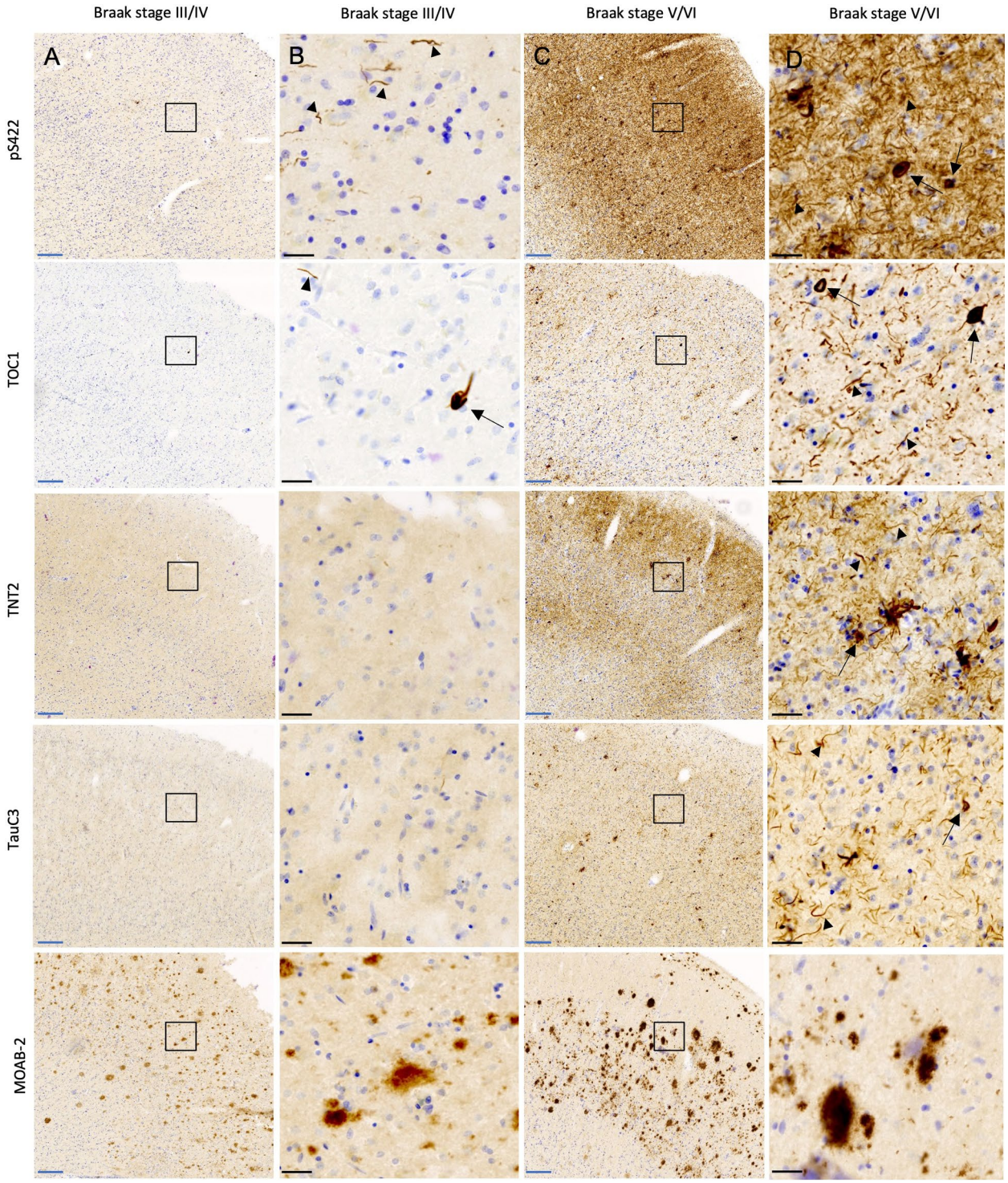


Fig. 1 Immunolabeling of early tau and amyloid pathology in PCC. Representative photomicrographs showing pS422, TOC1, TNT2, TauC3, and MOAB-2-immunoreactive profiles in layer 3/4 from a (**A**) Braak stage III/IV and (**C**) Braak stage V/VI case. Tau pathology was not apparent in the DMN hubs in Braak I/II cases. (**B**, **D**) High magnification micrographs showing labeling for each marker within the square boxes outlined in (**A**, **C**). Similar qualitative immunolabeling patterns were observed in FC and PreC. Scale bars: Blue = 200 µm, Black = 20 µm. Arrowheads = neuropil threads; Arrows = tau + somal inclusions

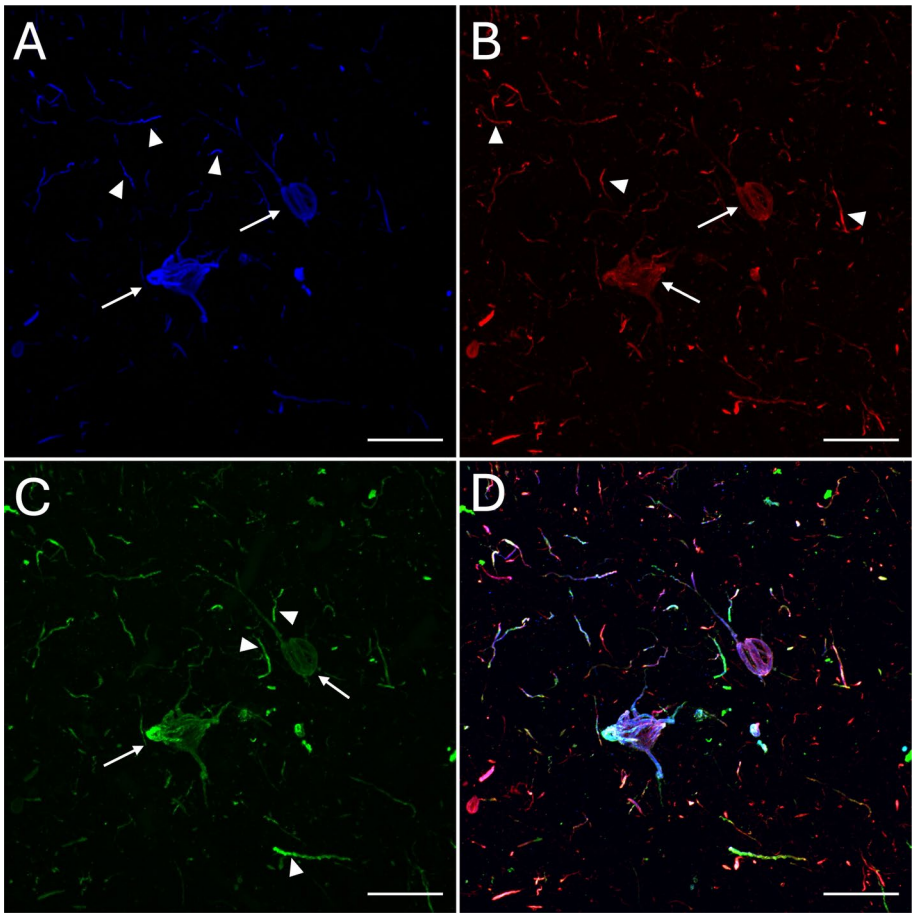


Fig. 2 Representative fluorescence micrographs of tau pretangle- immunoreactive profiles in PCC layers 3/4 of an AD case (Braak stage V). **A**) pS422 + profiles (blue), (**B**) TNT2 + profiles (red), and (**C**) TOC1 + profiles (green). **D**) Overlap of pS422, TNT2, and TOC1. Images were captured with a 40X oil objective. Scale bar: 25 µm. Note the differential expression patterns among the markers within neuropil threads (arrowheads), whereas marker overlap was more prominent in the neuronal soma (arrows)

higher in AD compared to NCI in FC ($p=0.0022$) and PCC ($p=0.0056$), but not PreC (Supplementary Fig. 3A). TNT2 and TauC3 levels were also significantly higher in AD compared to NCI; however, TNT2 levels in FC were significantly higher in MCI compared to NCI ($p=0.0271$) (Supplementary Fig. 3A). Levels of PHF-1 immunostaining were significantly increased in Braak stage VI and were significantly higher in AD compared to NCI (PCC, $p<0.0001$, Supplementary Fig. 4). Finally, MOAB-2 + amyloid load did not differ across diagnostic groups except for a significant increase in PreC in e AD (Supplementary Fig. 3A).

Correlations between DMN pretangle tau markers and clinical pathologic variables

Antemortem neuropsychological test scores assessing various cognitive domains were used to evaluate the relationship between regional DMN tau pathology and cognitive performance within a year of death [52]. Strikingly, increasing levels of pS422, TOC1, TNT2, and TauC3 + profiles in all three regions strongly correlated with poorer antemortem MMSE and GCS scores, as well as poorer performance on measures of episodic memory. By contrast, TNT2 and

TauC3 levels, but not pS422 or TOC1 levels, in FC and PreC were associated with working memory, whereas none of the marker levels in PCC were associated with working memory, a cognitive domain less associated with DMN [77] (Fig. 3B, D, F; Supplementary Table 3). By contrast, MOAB2 + amyloid load, correlated with pS422, TOC1, and TNT2 in all regions, but was only associated with lower episodic and semantic memory scores and GCS in the PCC (Fig. 3D, Supplementary Table 3).

Regarding postmortem neuropathological diagnostic criteria, increasing pS422, TNT2, and TauC3 levels were significantly associated with Braak stage, CERAD, and NIA-Reagan scores in the DMN hubs, these associations were weaker for TauC3 (Supplementary Fig. 5 A, C, E; Supplementary Table 6). TOC1 levels followed a similar pattern in PCC and FC, but did not correlate with the pathological diagnostic criteria in the PreC. MOAB-2 levels were significantly associated with CERAD and NIA-Reagan scores in the DMN hubs, but not with Braak stage in FC and PreC. Interestingly, increasing DMN pretangle markers were associated with increasing semi-quantitative estimates of global NFT and neuritic plaque burden, but not diffuse plaque burden (Supplementary Fig. 5 A, C, E; Supplementary Table 6).

Soluble pretangle tau markers in DMN hubs

We performed complementary custom sELISA studies to quantify TOC1, TNT2, and TauC3 in soluble DMN extracts as a complementary approach to measure the accrual of diffusible, presumably toxic tau moieties. Technical issues precluded optimization of a pS422 assay. We also quantified total tau in the fractions using the Tau 5 antibody (see Methods). We found that TOC1 oligomeric tau levels were significantly increased during the transition from Braak stage IV to V in all three hubs (FC, $p < 0.0001$; PCC, $p = 0.003$; PreC, $p = 0.0006$) (Fig. 4A, C, E; Supplementary Fig. 2B). Likewise, TNT2 levels were elevated in Braak V FC and PCC, and in Braak VI PreC ($p = 0.0223$, $p = 0.0001$, $p = 0.0462$, respectively) cases. TauC3 levels were also increased in DMN hubs in Braak stages V in the FC and PCC ($p = 0.0545$, $p = 0.0128$, respectively) and in Braak stage VI in PreC ($p = 0.031$). By contrast, Tau 5 total tau levels were only increased in Braak VI cases (Fig. 4A, C, E; Supplementary Fig. 2B). When comparing these markers across e clinical groups, soluble TOC1 levels were significantly increased in AD compared to MCI in all three hubs and compared to NCI in PCC and PreC and TNT2 showed a similar pattern in the FC and PCC (Supplementary Fig. 3B). TauC3 levels were increased between NCI and AD in PCC, whereas total tau5 levels did not differ significantly between clinical groups (Supplementary Fig. 3B).

Correlations between soluble DMN pretangle tau markers and clinical pathologic variables

Similar to histological results, inverse associations were observed between increasing levels of pretangle tau in soluble DMN fractions and poorer antemortem performance on neuropsychological tests. For example, TOC1 levels were correlated with poorer episodic memory in PCC ($r = -0.39$; $p = 0.00006$) and PreC ($r = -0.50$; $p = 0.0000001$) but not FC ($r = -0.18$; $p = 0.08$). Similarly, they were inversely correlated with semantic memory performance ((PCC ($r = -0.37$; $p = 0.0002$), PreC ($r = -0.42$; $p = 0.00002$), and FC ($r = -0.26$; $p = 0.009$) (Figs. 4B, D, F; Supplementary Tables 3 & 4). TNT2 level associations with cognitive performance were more variable across region and test, neither TOC1 nor TNT2 was associated with working memory. TauC3 correlated with MMSE, GCS, episodic memory, semantic memory, and working memory in PCC and PreC, whereas total tau (Tau5) did not reveal any correlations (Figs. 4B, D, F; Supplementary Table 4).

With respect to postmortem neuropathological diagnostic criteria, increasing TOC1, TNT2, and TauC3 levels, but not Tau5 levels, were significantly associated with Braak stage, CERAD, and NIA-Reagan scores in the DMN hubs (Supplementary Fig. 5B; Supplementary Table 7). TOC1, TNT2, and TauC3 levels were also positively correlated with semi-quantitative estimates of NFT density, neuritic plaque density, and neuritic plaque number. Tau5 levels were not associated with any of the measures (Supplementary Fig. 5B; Supplementary Table 7).

(See figure on next page.)

Fig. 3 Pathological tau accumulation in the DMN across Braak stage and correlations with antemortem cognitive performance. A, B) pS422, TOC1, TNT2, TauC3, and MOAB2 levels (A) and correlations of tau marker levels with antemortem cognitive assessments (B) in FC. C, D) Tau and amyloid pathology marker levels (C) and correlations (D) for PCC. E, F) Tau and amyloid pathology marker levels (E) and correlations (F) for PreC. % area values were log-transformed only for visualization purposes and compared via Kruskal–Wallis with Dunn's correction. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$. Bar graphs show mean ± SD. Spearman correlations tested associations between marker levels and clinical pathologic variables (*rho* coefficients are shown), *, significant correlations (see Supplementary Tables 3 and 6 for *p* values). Abbreviations: MMSE, Mini-Mental State Exam; GCS, Global Cognitive Score; EM, Episodic memory; SM, Semantic memory; WM, Working memory; PS, Perceptual speed; VA, Visuospatial ability

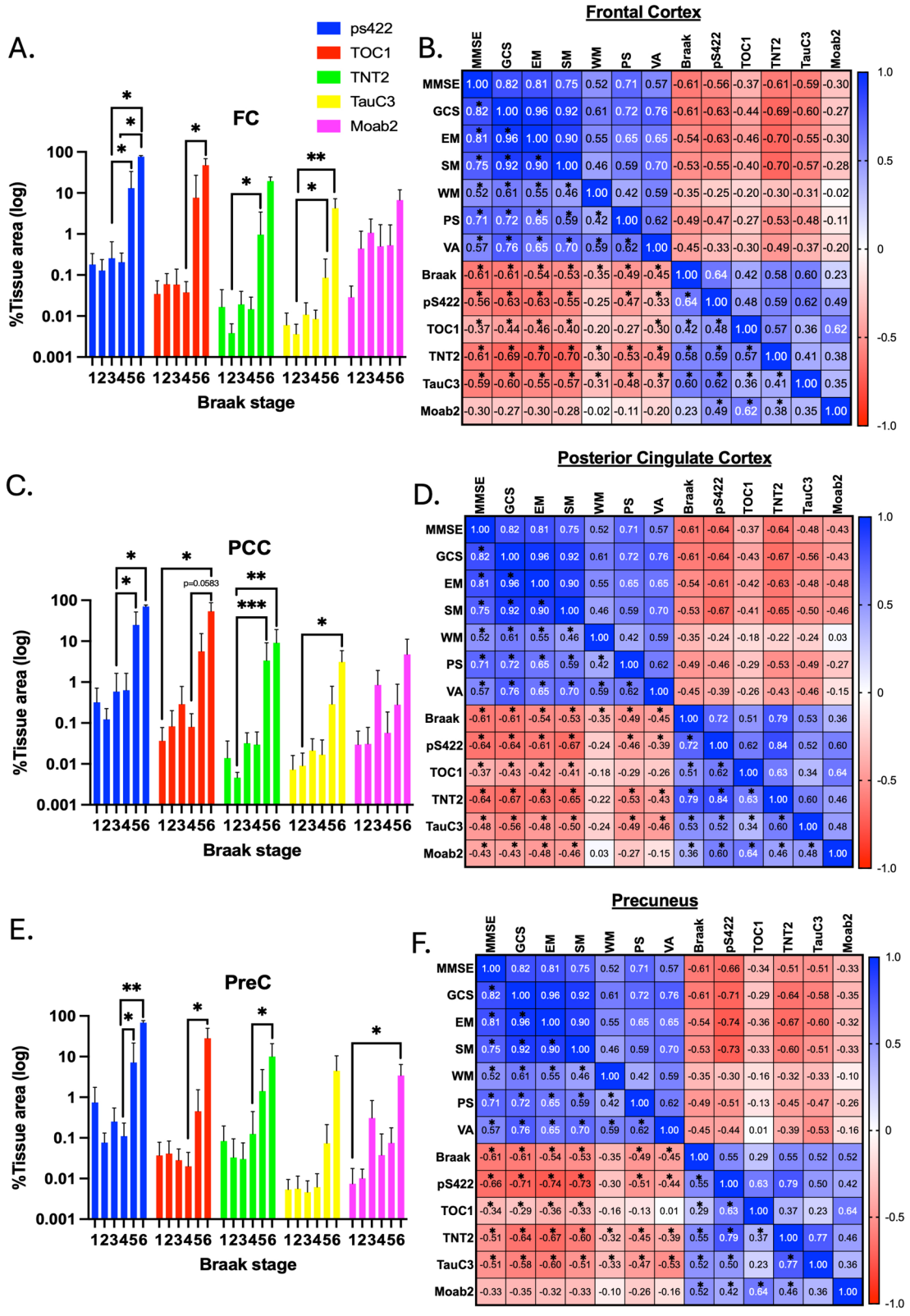


Fig. 3 (See legend on previous page.)

Validation studies in MADC subjects

DMN samples from the MADC clinical cohort were used to validate the soluble tau sELISA finding derived from RROS tissue (Table 3). Subjects were classified as low-Braak/control (Braak stage I-II), mid-Braak (III-IV), and high-Braak (V-VI) and did not differ in age, sex, or PMI across groups. However, MMSE scores were significantly lower in the high Braak group (Table 3). Interestingly, regional soluble tau pathological load was higher for all markers in successive Braak stages in this cohort (Fig. 5A, C, E) compared to the RROS (Fig. 4). Moreover, while soluble TOC1 and TNT2 levels were significantly increased during the transition from Braak stage IV to Braak stage V in the DMN hubs in RROS cases, these markers were increased during the transition from Braak stage III to Braak stage IV in the DMN hubs in the MADC cases (Fig. 5). However, except for TOC1 in PCC, significance was not reached until Braak stage V, similar to the RROS cohort. TauC3 load was minimal in PCC and PreC, yet showed higher abundance in FC. There were no differences in Tau5 + total tau levels across Braak stages. Finally, a strong and statistically significant inverse correlation was observed between soluble TOC1 and TNT2 levels and antemortem MMSE scores across all three DMN regions (Figs. 5B, D; Supplementary Table 5). In contrast, TauC3 levels correlated with MMSE scores only in PreC, similar to the RROS cohort (Fig. 5F).

Distribution of cognitive scores across Braak stage in the RROS cohort

MMSE and GCS scores were plotted by Braak stage for each RROS subject (Fig. 6). Both scores of antemortem global cognitive performance were significantly decreased between Braak stages IV and V, mirroring the pattern of significant increases in pretangle tau markers across Braak stages (Figs. 3 and 4).

Regional differences in DMN tau pathology in the RROS cohort

When comparing regional distributions of the tau markers within DMN hubs, pS422 was the most abundant marker and its expression levels were significantly higher in the PCC ($p = 0.0029$) compared to FC and PreC (Fig. 7). Similarly, both total TOC1 and soluble TOC1 were higher in the PCC. Total TNT2 was also higher in the PCC compared to the FC. Soluble TauC3 was slightly higher in the FC compared to the PreC. Finally, MOAB-2 was significantly higher in the PCC and FC compared to PreC. The overall pattern indicated divergence in the posterior DMN hubs, with a higher pre-tangle tau load in the PCC and possible resistance to tau and amyloid pathology in neighboring PreC.

Discussion

Tau epitopes associated with pretangle cellular inclusions are linked to impairment of cellular trafficking, ionic imbalances through disrupting ion-gated calcium channels, and reduced mitochondrial and lysosomal functions [11, 78–80]. These, and likely other cellular disturbances, result in synaptic impairments (e.g., disrupted LTP) leading to neuronal death, synaptic disconnection, and cognitive dysfunction [14, 17, 81, 82]. However, the threshold accumulation of tau toxic pretangle pathology underlying cortical dysfunction required to precipitate cognitive impairment remains unknown. Here, we investigated pretangle cellular and neuropil tau pathology in the FC, PCC, and PreC hubs of the DMN connectome that mediates memory and attention function and is dysregulated in the preclinical and prodromal stage of AD [83–85]. To our knowledge, this is the first study to examine pretangle tau pathology across hubs of the DMN using tissue obtained from cases with an antemortem clinical and postmortem neuropathological evaluation. To accomplish this, we used multiple site-specific tau antibodies to reveal pretangle lesions within hubs of the DMN across clinical diagnostic categories and during the onset of Braak stages. We found a step-like increase in DMN tau pathology across clinical diagnostic groups, showing a significant increase in AD compared to NCI, with intermediate levels in MCI. In addition, pretangle tau markers correlated strongly with measures of episodic and semantic memory, global cognitive scores, and Braak, CERAD, and NIA-Reagan scores,

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Fig. 4 Pathological tau accumulation in DMN soluble fractions across Braak stage and correlations with antemortem cognitive measurements. TOC1, TNT2, TauC3, and Tau5 levels and correlations of tau marker levels with antemortem cognitive assessments (**B**) in FC. **C**, **D**) Tau marker levels (**C**) and correlations (**D**) for PCC. **E**, **F**) Tau marker levels (**E**) and correlations (**F**) for PreC. % Light absorbance was compared via Kruskal–Wallis with Dunn's correction. *, $p < 0.05$; ***, $p < 0.001$; ****, $p < 0.0001$. Bar graphs show mean ± SD. Spearman correlations tested associations between marker levels and clinical pathologic variables (*rho* coefficients are shown), *, significant correlations (see Supplementary Tables 4 and 7 for p values). Abbreviation**ns:** MMSE, Mini-Mental State Exam; GCS, Global Cognitive Score; EM, Episodic memory; SM, Semantic memory; WM, Working memory; PS, Perceptual speed; VA, Visuospatial ability

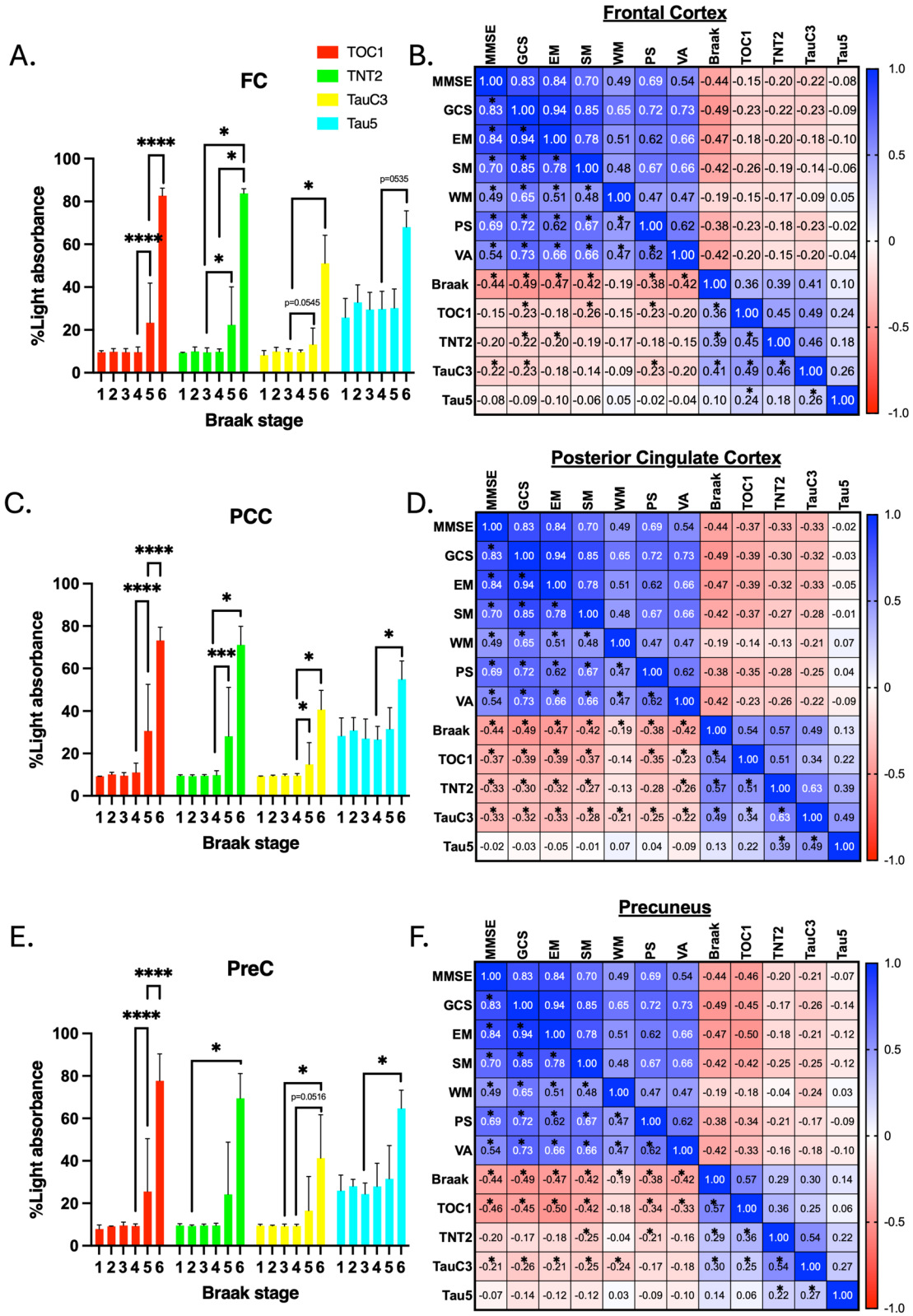


Fig. 4 (See legend on previous page.)

Table 3 Demographic, Clinical, and Pathological Profile of the MADC cohort

	Clinical diagnosis				Comparison by diagnosis group
	low-Braak	mid-Braak	high-Braak	Total	(*P* value)
	(*n* = 12)	(*n* = 12)	(*n* = 12)	(*n* = 36)	
Age at death (years)					
Mean ± SD	67.8 ± 14.9	80.3 ± 10.8	74.3 ± 10.5	74.1 ± 7.6	0.056[a]
(Range)	(39–87)	(60–95)	(57–92)	(39–95)	
No. (%) Males	6 (50%)	6 (50%)	6 (50%)	18 (50%)	> 0.9999[‡]
Education					
Mean ± SD	15.25 ± 2.2	15.92 ± 2.4	15.33 ± 2.8	15.6 ± 2.5	0.8376[a]
(Range)	13–18	12–20	12–20	12–20	
Postmortem Interval (hours)					
Mean ± SD	12.7 ± 7.6	8.6 ± 5.3	9.6 ± 5.5	10.2 ± 6.3	0.298[b]
(Range)	(4–26)	(3–18)	(4–21)	(3–26)	
MMSE					
Mean ± SD	27 ± 2	15.3 ± 11.9	8.1 ± 6.9	14.4 ± 10.9	0.0143 [b]
(Range)	(24–28)	(1–29)	(0–18)	(0–29)	
Braak Scores					
I-II	12	0	0	12	< 0.0001[b]
III-IV	0	12	0	12	
V-VI	0	0	12	12	

[a] One-way ANOVA and Tukey's correction

[b] Kruskal–Wallis and Dunn's correction

[‡] Chi-Square test

MMSE Mini-Mental State Exam, *SD* Standard deviation

as well as semi-quantitative measures of NFT and neuritic plaque burden.

Pathological tau quantification in the DMN

Clinical pathological studies of tau in AD have mainly focused on NFTs in the trans/entorhinal cortex and hippocampal components of the medial temporal lobe memory circuit, which occur prior to the spread of NFT into neocortical hubs of the DMN. Quantification of early pathological pretangle tau phosphorylation and conformational epitopes revealed that pS422 (Serine 422 phosphorylation) was the earliest and the most abundant marker that primarily labeled NTs prior to appearing in the cell soma suggesting an early axonal transport defect similar to seen in association with the cholinergic basal forbrain cortical projection neurons [63, 86, 87], entorhinal cortex pyramidal neurons [64] and in synaptosomes isolated from the angular gyrus [88] during the progression of AD. Phosphorylation of tau at S422 residue is associated with several kinases, including GSK3B, JNK3, and p38 [13, 62, 87, 90], which are associated with tau toxicity through various molecular mechanisms [91–94]. S422 phosphorylation also inhibits anterograde and retrograde fast axonal transport [13], disrupts protein homeostasis [86], and is linked to neurotransmitter abnormalities [95]. Recently,

(See figure on next page.)

Fig. 5 Pathological tau accumulation in DMN soluble fractions in MADRC subjects and correlations with MMSE scores and Braak stage. TOC1, TNT2, TauC3, and Tau5 levels and correlations of tau marker levels with MMSE and Braak stage in (**A**, **B**) FC, (**C**, **D**) PCC, and (**E**, **F**) PreC. % Light absorbance was compared via Kruskal–Wallis with Dunn's correction. *, $p < 0.05$; **, $p < 0.01$. Bar graphs show mean ± SD. Spearman correlations tested associations between marker levels and MMSE score or Braak stage (*rho* coefficients are shown), *, significant correlations (see Supplementary Table 5 and 8 for p values). Abbreviations: MMSE, Mini-Mental State Examination; GCS, Global Cognitive Score; EM, Episodic memory; SM, Semantic memory; WM, Working memory; PS, Perceptual speed; VA, Visuospatial ability

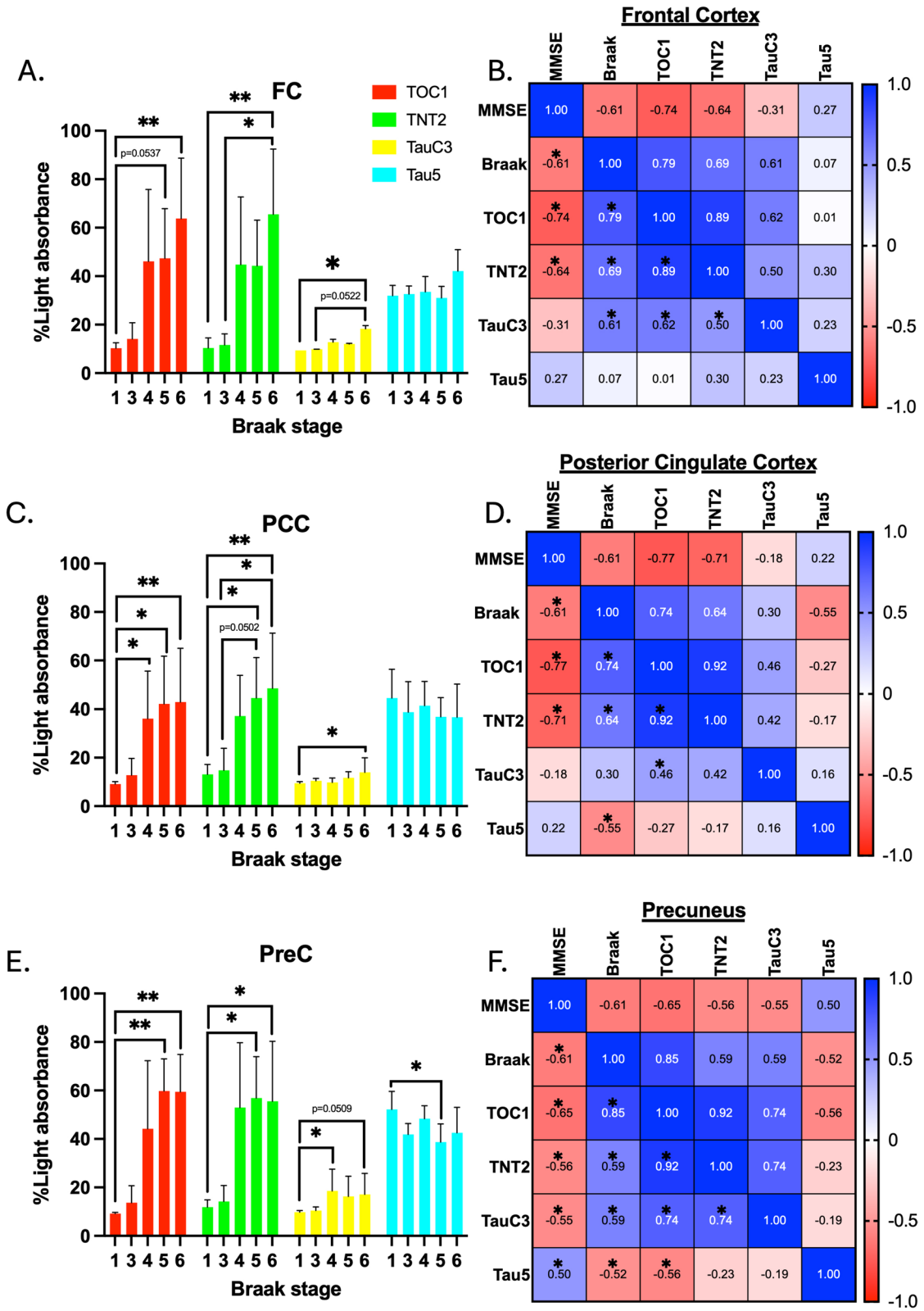


Fig. 5 (See legend on previous page.)

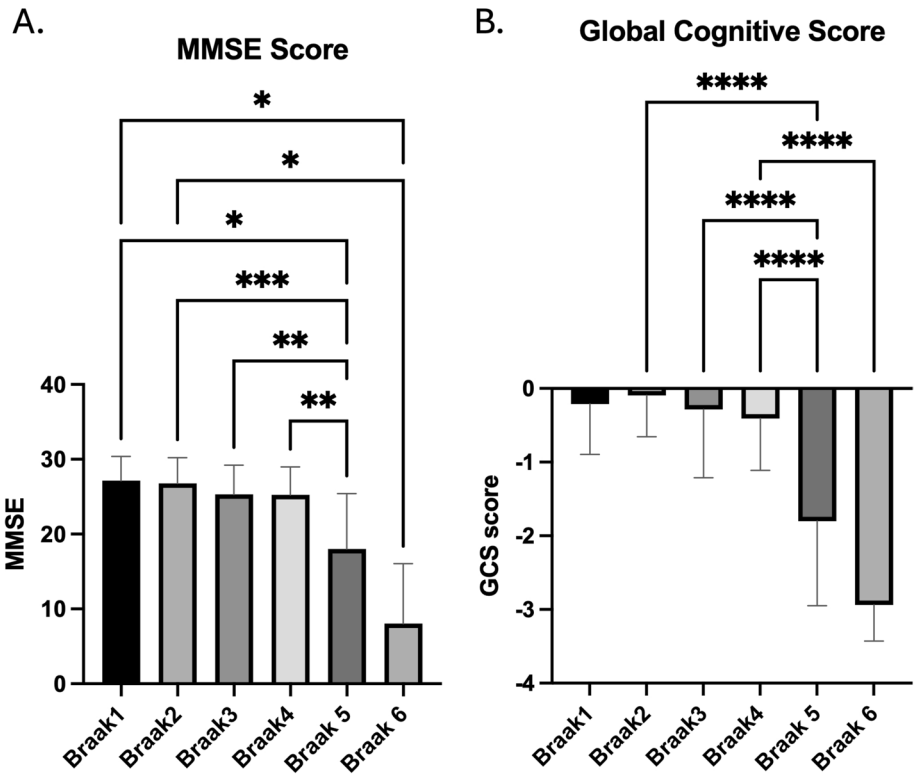


Fig. 6 Distribution of global cognitive scores across Braak stage in the RROS cohort. **A**) Antemortem MMSE scores for RROS subjects distributed across their individual postmortem Braak stage diagnosis, via Kruskal–Wallis and Dunn's correction. **B**) Antemortem GCS for each RROS subject across Braak stage, via one-way ANOVA and Tukey's correction. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$. Bar graphs show mean ± SD. Abbreviations: GCS, Global Cognitive Score; MMSE, Mini-Mental State Exam

Stefanoska and colleagues investigated Serine/Threonine phosphorylation sites of tau in vitro, identifying S422 as a key residue in the C-terminus of tau, where its phosphorylation modulates the phosphorylation of other tau epitopes interdependently [96], . In addition to the DMN, oligomeric (TOC1-positive [65]) and N-terminal misfolded (TNT2-positive [75]) tau are present in the entorhinal cortex and medial temporal lobe [66, 97–99] as well as the frontal cortex in other tauopathies [89, 100]. Previously, TOC1-positive tau levels were found to increase in synaptosome-enriched fractions from layer II of the visual cortex in Braak stage III/IV cases [101]. Similar to pS422, TOC1 + and TNT2 + NTs were largely confined within the cortical laminae of the DMN hubs during early Braak stages (Figs. 1, 2) with the gradual appearance of neuronal labeling in layers 2/3 and 5/6 by Braak stage V (Fig. 1) suggesting that these pathological profiles underlie the significant increase in pretangle tau burden that occurs during the transition from Braak stage IV to V in the DMN. The abundance of TOC1 and TNT2 followed a similar pattern; however, their structural localization differed slightly. TNT2 staining was more pronounced than TOC1 in NTs, whereas TOC1 appeared early in the cell soma (Fig. 1**,** Supplementary Fig. 1A).

Although the origin of these neurites remains unclear, perhaps they arise from corticocortical or thalamocortical projection neurons [102]. Moreover, the DMN hubs also receive extensive innervation from cholinergic basal forebrain [103, 104] and brainstem noradrenergic [105, 106] and serotinergic [107] neurons that modulate memory and attentional behaviors [108]. Although the process(es) underlying the onset of NT pathology remains unknown, the extensive and early development of NTs reported during the evolution of tau pathology within the cholinergic basal forebrain suggests a defect in retrograde axonal transport [63]. Whether this is a unifying concept for the development of NTs during the onset of AD remains a major area of interest.

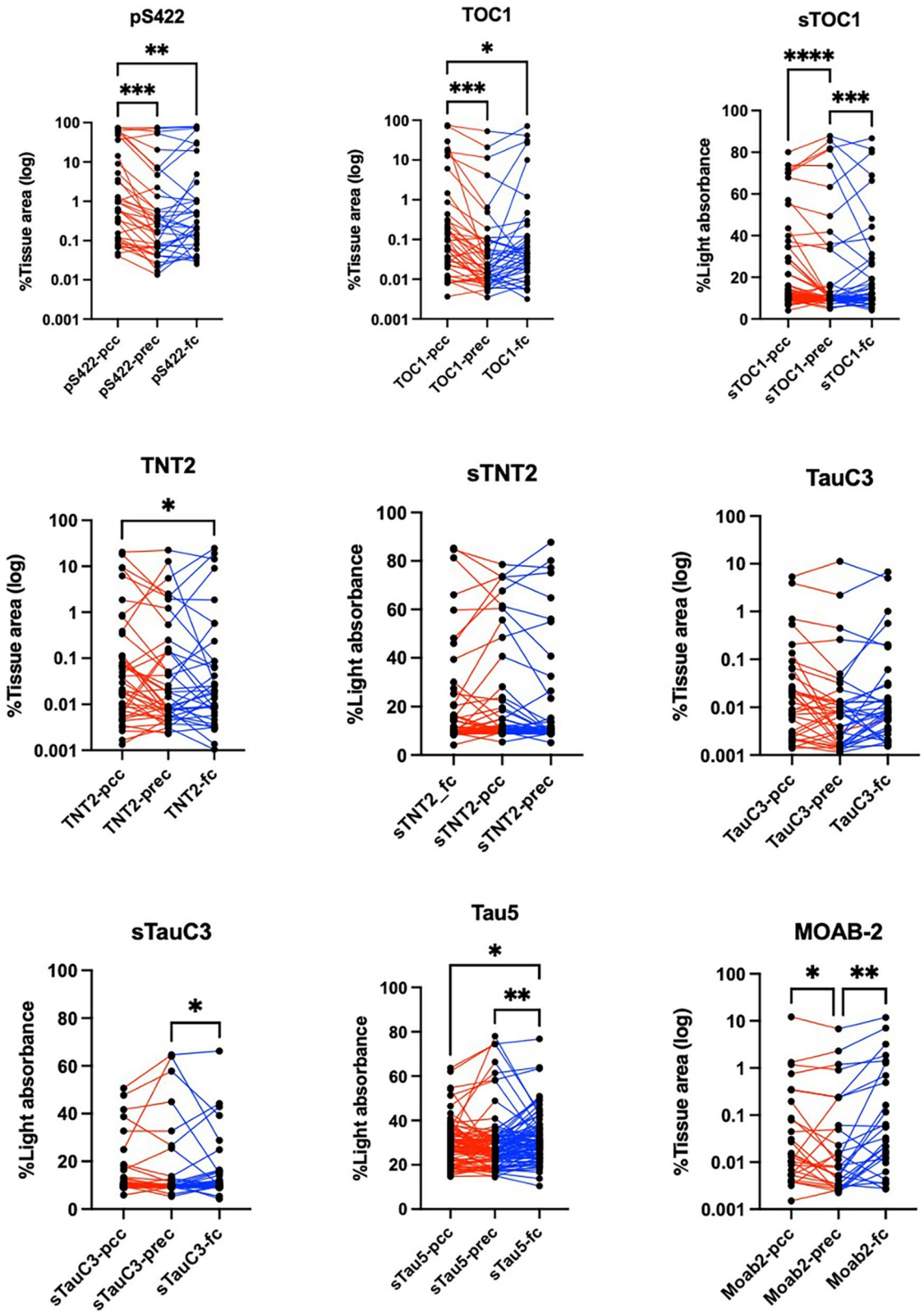


Fig. 7 Regional comparisons of soluble and total tau markers. Regional differences in the abundance of selected markers across regions. IHC and ELISA-measured (i.e., sTOC1) markers in the PCC, PreC, and FC were paired for each case via the nonparametric Friedman test with Dunn's correction. s = measured in the S1 soluble fraction. *, $p < 0.05$; **, $p < 0{,}01$; ***, $p < 0{,}001$; ****, $p < 0.0001$

The midstage marker TauC3 detects the truncation of aspartic acid at residue 421 is achieved by caspase-mediated proteolysis [67, 68], which inhibits axonal transport [109]. In the current study, the appearance of TauC3-labeled profiles was more similar to TOC1, with relatively less prominent immunopositive NTs and more neuronal enrichment within the DMN hubs (Fig. 1). TauC3 pathology was also less abundant compared to the other markers in the early Braak stages, suggesting its later appearance is downstream of potential pretangle-mediated neurodegenerative processes resulting in a pro-apoptotic cellular environment.

DMN tau correlates with episodic and semantic memory

We previously showed that pS422 and TOC1 colocalize within cholinergic nucleus basalis neurons, where their expression levels correlate with GCS and MMSE scores [110]. Consistent with these findings, we found strong inverse correlations between select pathological tau markers in the DMN and GCS and MMSE scores, particularly episodic and semantic memory. Episodic memory is a type of explicit long-term memory that involves recalling past events and experiences and is primarily encoded, stored, and processed within the medial temporal lobe memory circuit and the PCC and PreC hubs of the DMN [111–114]. Episodic memory decline is a clinical hallmark of AD onset [115] but also declines with normal aging [116, 117]. Semantic memory, which involves the acquisition of facts, concepts, and general knowledge throughout life, remains intact with age, but declines during the amnestic MCI prodromal stage ofAD [113, 118, 119]. Semantic memory has been suggested to be a potential clinical marker differentiating AD from pathological aging or other neurodegenerative diseases [22, 120]. Additionally, pretangle tau pathology correlates to varying degrees with perceptual speed and visuospatial ability, which are mediated by the DMN connectome [121, 122]. By contrast, the role of DMN function in working memory is debatable [77, 123]. The lack of association between DMN pretangle tau burden and working memory performance suggests that this cognitive domain is either resilient to tau neurodegeneration or that the DMN plays a limited role in working memory function. However, the present findings underscore potential roles that the accumulation of early pathological tau species plays in episodic and semantic memory during the onset of AD. These data also highlight the potential of the DMN as a model paradigm for studying mechanistic clinical pathologic relationships in the context of this vulnerable connectome in AD.

Oligomeric amyloid pathology in the DMN

The bidirectional interaction between Aβ and tau pathology in AD is well recognized, with evidence of co-pathological exacerbation of proteinopathy and neuronal dysfunction [124–126]. In this regard, previous PET and fMRI studies showed that amyloid plaque positivity in the DMN is associated with hyperconnectivity [40]. However, DMN functional connectivity shifts toward a state of hypoconnectivity once NFT co-pathology is detectable by PET. More recently, combined PET/fMRI studies in two separate cohorts showed that Aβ-associated hyperconnectivity may modulate the accumulation of tau pathology in the same connected brain regions, providing a potential link between Aβ deposition and NFT-associated resting state network failure [127]. We also immunolabeled oligomeric Aβ using the MOAB-2 antibody and examined the correlation between oligomeric Aβ and tau pathology, as well as cognitive scores. Interestingly, MOAB-2 levels were strongly associated with pS422, TOC1, and TNT2 in all three DMN regions, but oligomeric amyloid burden generally did not associate with cognitive deterioration except in the PCC (Fig. 3, Supplementary Table 3) Together, these findings support the prevailing concept that tau pathology more strongly impacts cognitive decline than amyloid pathology, even within the DMN connectome. Further studies are needed to validate these results and explore the complex interplay between these co-pathologies.

Soluble tau pathology within the DMN

Quantitative morphometric assessments were complemented by ELISA evaluations focusing on the accumulation of soluble pretangle tau monomers and oligomers in the DMN. Similar to morphologic pathology, tau levels significantly increased from Braak IV to V. However, weaker associations were found between TOC1, TNT2, and TauC3 levels with antemortem cognitive measures, including MMSE and GCS, in the soluble protein fraction compared to structural assessments, including NT and NFT labeling (Figs. 3B, D, F & 4B, D, F). This observation is reminiscent of studies demonstrating that cognitive deterioration accelerates with the appearance of NFTs in higher association neocortical regions [128], suggesting that a threshold level of soluble toxic tau drives NFT pathology. In support of this concept, we showed that poorer antemortem global cognition was not significantly decreased until the transition from the limbic Braak stage IV to neocortical Braak stage V (Fig. 6) in the cases from the RROS cohort.

Interestingly, our validation cohort using postmortem DMN soluble fractions from MADC subjects revealed a similar abrupt increase in pretangle tau pathology with increasing Braak stage, but this increase occurred between Braak stages III and IV instead of IV and V, as seen in the RROS cases. A possible explanation for this may relate to differences between the cohort subjects and

lifestyle factors. The RROS cohort, consisting of retired Catholic clergy, benefited from more years of formal education (Tables 1, 5) and a relatively healthy lifestyle, including a more regulated diet, structured sleep patterns, medical care, and religious activities that may provide cognitive benefits as well as stress reduction [52]. Additionally, the strong sense of community fostered by cloistered living both promotes the beneficial effects of lifelong learning and prevents social isolation, which are modifiable risk factors for dementia [129, 130]. In contrast, the MADC cohort represents a community-dwelling population, with more potential exposures to metabolic, vascular, epigenetic modulation, and transcriptional regulation of the genome [49, 131, 132]. The early accrual of tau pathology despite the MADC subjects having a younger average age (74.1 ± 7.6) at death, suggests that the RROS cohort was enriched for pathological resistance and cognitive resilience [133]. Alternatively, the observed differences could be due to minor differences in neuropathological interpretation or regional brain dissection between the two cohorts.

Regional differences in tau among the DMN hubs

Our findings that tau pathology was highest in the PCC align with previous reports that the PCC is involved very early in AD progression, possibly due to an early decline in its high metabolic baseline determined by FDG-PET [46, 134]. Interestingly, Aβ initiation and accumulation rates were highest in PCC compared to other brain regions, with PreC displaying relatively delayed amyloid and tau pathology, suggesting resistance [135]. The proposed resistance of PreC could be due in part to the preservation of neurotrophic signaling pathways despite accumulating Aβ pathology [136], differences in gene regulation within PreC pyramidal neurons compared to the other DMN hubs [137], or differences in splicing protein mislocalization between the PCC and PreC [138].

Study limitations

The present study has limitations. First, the outcome measures used machine-learning software to quantify the immunolabeled images based on the percentage of labeled tissue, which allowed the normalization of the signal relative to the area of the region of interest. However, this may result in an underestimation of total tau pathology. For instance, the predominance of NTs seen in the current microscopy studies appeared first in early Braak stages in the DMN hubs, but did not coincide with a high densitometric signal per region of interest. Hence, the extent of NT pathology may be more accurately measured using stereological counting techniques [139].

Secondly, PET and fMRI sequences are not performed at the time of annual clinical evaluations in the RROS cohort. This precluded comparing postmortem tau profiles with tau-PET and DMN functional connectivity prior to death. On the other hand, detailed postmortem records of NFT and plaque pathology were available, including plaque density, plaque number, and NFT counts in key regions like the entorhinal cortex, hippocampus, and cortical regions. These data were incorporated into the correlation matrices, providing valuable insights into the relationship between pathologies.

Thirdly, whether the development of novel biofluid or neuroimaging biomarkers for nonfibrillar oligomers will enhance early diagnosis and currently approved amyloid targeting drugs combined with future tau therapies to treat AD remains an ongoing area of research [140]. In this regard, it is possible that the present data describing the involvement of tau and amyloid oligomers during the clinical and pathological onset of AD will contribute to this endeavor.

Finally, as AD has been suggested to be a multi-system disorder [141] affecting various biological systems and processes, NFT pathology often overlaps with other conditions, such as varying degrees of arteriolosclerosis and cerebral amyloid angiopathy (CAA) [142]. In this study, some subjects were positive for CAA, which may have exacerbated cognitive decline in addition to the coexisting tau pathology.

Conclusions

The DMN, which displays hallmark AD pathology early in the onset of the disease, is a excellent candidate to study clinical pathologic correlations and potential mechanisms of tau-mediated neurodegeneration across tauopathies [42, 143, 144]. Increased NFT tracer retention shows strong correlations with inter- and intra-functional connectivity across DMN hubs in combined fMRI/PET studies [24, 40, 134]. However, pretangle tau accumulation within the DMN during AD progression has never been explored systematically. Here, we showed for the first time that pretangle tau markers were present at low Braak stages, predominantly as NTs, and significantly increased at Braak stages IV to V. The markers increased in the DMN in a step-wise manner from NCI to MCI to AD and were strongly correlated with antemortem measures of cognitive deterioration, particularly in episodic and semantic memory, as well as additional cognitive domains mediated by the DMN including perceptual speed and visuospatial attention. These findings contribute to our understanding of the the timeline of AD pathophysiology within a key, vulnerable resting state network, and suggest that therapeutics aimed at pretangle forms of tau may preserve brain connectome functionality in the DMN.

Abbreviations

Aβ	Beta Amyloid
AD	Alzheimer's Disease
ANOVA	Analysis of Variance
APOE	Apolipoprotein E
BCA	Bicinchoninic Acid
CAA	Cerebral Amyloid Angiopathy
CERAD	Consortium to Establish a Registry for Alzheimer's Disease
DMN	Default Mode Network
ELISA	Enzyme-Linked ImmunoSorbent Assay
FC	Frontal Cortex
fMRI	Functional Magnetic Resonance Imaging
GCS	Global Cognitive z-Score
GSK3β	Glycogen Synthase Kinase-3 Beta
HRP	Rabbit Horseradish Peroxidase
IF	Immunofluorescent
IHC	Immunohistochemistry
JNK3	c-Jun N-terminal Kinase 3
LED	Light-Emitting Diode
LTP	Long Term Potentiation
MADC	Michigan Alzheimer's Disease Center
MCI	Mild Cognitive Impairment
MMSE	Mini-Mental State Examination
NCI	No Cognitive Impairment
NFT	Neurofibrillary Tangle
NT	Neuropil Threads
PAD	Phosphatase Activating Domain
PCC	Posterior Cingulate Cortex
PET	Positron Emission Tomography
PMI	Postmortem Interval
PreC	Precuneus
RROS	Rush Religious Orders Study
SCD	Subjective Cognitive Decline
SD	Standard Deviation
TDP-43	Transactive Response DNA-Binding Protein 4

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s44477-026-00019-y.

Below is the link to the electronic supplementary material.Supplementary file1 Co-labeling with pS422, TOC1, and TNT2/TauC3 in PCC layers 3 and 4 of an AD case (Braak stage V) A) Immunofluorescence co-labeling of pS422, TNT2, and TOC1. B) Immunofluorescence co-labeling of pS422, TNT2, and TauC3. Note that pS422, TOC1, and TNT2 displayed significant overlap, particularly in the neuronal soma, whereas TauC3 epitopes displayed minimal overlap. Image captured with a 40X oil objective. Scale bar: 25 µm. Arrowheads = neuropil threads; Arrows = tau+ somal inclusions (PDF 140 KB)

Supplementary file2 DMN pretangle tau distribution across RROS cases via Braak staging. A) Tau and amyloid pathology markers quantified via IHC. B) Tau pathology markers quantified in soluble fractions via ELISA, via Kruskal Wallis with Dunn's correction. *, $p < 0.05$; ***, $p < 0,001$; ****, $p<0.0001$. Abbreviations: FC, frontal cortex; PCC, posterior cingulate cortex; PreC, precuneus (PDF 433 KB)

Supplementary file3 DMN pretangle tau distribution across RROS cases via clinical diagnostic group. A) Tau and amyloid pathology markers quantified via IHC. B) Tau pathology markers quantified in soluble fractions via ELISA, via Kruskal Wallis with Dunn's correction. *, $p < 0.05$; **, $p < 0.01$ ***, $p < 0,001$. Abbreviations: NCI, no cognitive impairment; MCI, mild cognitive impairment; AD, Alzheimer's disease; FC, frontal cortex; PCC, posterior cingulate cortex; PreC, precuneus (PDF 450 KB)

Supplementary file4 PHF-1 load in PCC and clinical pathologic correlations in the RROS cohort. A) PHF-1 levels based upon Braak stage (left) or clinical diagnostic group (right), via Kruskal-Wallis with Dunn's correction. B) PHF-1 associations with antemortem cognitive assessment, via Spearman correlation. *, $p < 0.05$; **, $p < 0,01$; ****, $p < 0.0001$ Abbreviations: AD, Alzheimer's disease; GCS, global cognitive z-score; MCI, mild cognitive impairment; NCI, no cognitive impairment; PCC, posterior cingulate corte; MMSE, Mini-Mental State Examination; GCS, Global Cognitive Score; EM, Episodic memory; SM, Semantic memory; WM, Working memory; PS, Perceptual speed; VA, Visuospatial ability (PDF 72 KB)

Supplementary file5 Correlations between the pretangle DMN tau levels and neuropathological diagnostic scores in the RROS cohort. Correlation coefficients of the Spearman correlations for the IHC (Supplementary Figure 5 A, C, E) and ELISA (Supplementary Figure 5 B, D, F)-measured selected markers and the postmortem neuropathological scores, such as neuritic plaque, diffuse plaque, TDP-43, and CAA scores in each region. TDP-43: TAR DNA-binding protein 43, CAA: Cerebral Amyloid Angiopathy. Heat map reflects the strength of correlation (PDF 200 KB)

Supplementary file6 (PDF 86 KB)

Supplementary file7 (PDF 112 KB)

Supplementary file8 (XLSX 15 KB)

Supplementary file9 (XLSX 14 KB)

Supplementary file10 (XLSX 10 KB)

Supplementary file11 (XLSX 17 KB)

Supplementary file12 (XLSX 15 KB)

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Author contributions
SEC conceived of and designed the research project with intellectual input from EJM, IEV, NKM, and BK. BK performed the histological and biochemical studies with critical assistance from IMK and JSB. Statistical analysis was performed by BK, NK, and SEC. The manuscript was written by BK and SEC with substantive input from EJM, IEV, and NMK. All authors approved the manuscript.

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Data availability
The datasets analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
All subjects provided signed informed consents for clinical evaluations and brain donations that were ethically approved by the Rush University Medical Center Institutional Review Board.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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